Physics Concepts as Indicative Framework to Model Systems of Economies Ex-ante

Coincise Notes and Conjectures Explorations of Natural Laws as Approach to Preliminarly Structure Economies - A Distillation of Corroboration of Mathematical Models

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An exploratory attempt … Interconnection network between movement, force, momentum, energy, and work. Describing systems with Lagrangian and Hamltonian when constraints and starting conditions exist. Modeling systems and optimisation with constraints and conditions. Interwoven system of macro- and microeconomie under constraints, limitations, and conditions.

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WORKING NOTES DRAFT

POTENTIALS

Potential as "intrinsic energy of inertia"

Electro-Magnetic Field (force caused by potential)

- static: lovslized source charge at rest, only spacial dependency, no time dependency
- stationary: source charce at uniform movement, steady current flow, space and time dependency
- dynamic: source charge at fast movement, radiation, space and time dependency
- relativistc: source charge at very speedy movement, radiation, space and time dependency

- rest localization vs. time-dependency
- slow vs. fast movemnt
- vacuum vs. matter
- local vs. global of $\mathcal{B}$, singularity vs. regularity of $\mathcal{E}$
- macro vs. micro

- far-field vs. near-field

Magnetic Field in Vacuum

$$\mathcal{B} = \nabla \times U_m := rot(A) \overset{\text{Matter}}{=} \mu_0 \left(\mathcal{H} + \mathcal{M}\right) \overset{\text{Vacuum}}{=} \frac{1}{c}\left(\frac{r}{|r|} \times \mathcal{E}\right) \overset{\text{Far Field}}{=} \frac{1}{2}\frac{1}{2\pi}\mu_0 \frac{k^2 e^{i(kr-\omega t)}}{r}\frac{r}{|r|} \times \left(cp_{0_{dipol}} + m_{0_{\cdot}} \right. \tag{1}$$

Magnetic Field in Matter

Exitation $\hat{=}$ Displacement

$$\text{Excication:} \qquad \mathcal{H} = \frac{1}{\mu_0}\mathcal{B} - \mathcal{M} = \frac{1}{\mu}\mathcal{B} \quad \text{Magnetisation:} \qquad \mathcal{M} = \frac{\mu - \mu_0}{\mu\mu_0}\mathcal{B} = \frac{\mu - \mu_0}{\mu_0}\mathcal{H} \tag{2}$$

Electric Field in Vacuum

$$\mathcal{E} \overset{\text{conservativ}}{=} -\nabla U_e \overset{\text{dynamic}}{=} -\nabla U_e - \frac{\partial}{\partial t}U_m \overset{\text{fast moving charge}}{=} \frac{1}{2}\frac{1}{2\pi}\frac{1}{\varepsilon_0}\frac{1}{r}\frac{1}{\left(1 - \frac{r}{|r|}\cdot\frac{v}{c}\right)^3} q \left[\underbrace{\left(1 - \frac{v^2}{c^2}\right)\frac{\frac{r}{|r|} - \frac{v}{c}}{r}}_{\text{uniform movement}} + \underbrace{\frac{\frac{r}{|r|} \times \left[\left(\frac{r}{|r|}\right.\right.}{\cdot}}_{\text{accelerate}} \right. \tag{3}$$

Electric Field in Matter

Displacement $\hat{=}$ Exitation (field change)
dielectricum = non-conducting material
electricum = conductor medium

$$\text{Excication:} \qquad \mathcal{D} = \varepsilon_0\mathcal{E} + \mathcal{P} = \overset{\text{isotropic medium in weak fields}}{=} \varepsilon_0\mathcal{E} + \varepsilon_0\chi\mathcal{E} = \varepsilon_0\left(1 + \chi\right)\mathcal{E} = \quad \varepsilon_0 4\pi r^3 n \left\langle p_m \right\rangle = 4\pi r^3 n \tag{4}$$

Abbreviations

$\Phi =$ electric potential $\quad A =$ magnetic potential $\quad j = \nabla \times \mathcal{H} =$ elec. current density

$\mathcal{E} =$ Electric Field $\quad \mathcal{B} =$ Magnetic Field $\quad \nabla\cdot\mathcal{B} = 0 \quad \mathcal{D} =$ Displacement density of dielectricum $\quad \mathcal{M} =$

$\mathcal{P} =$ Polarisation $\quad \mathcal{H} =$ magn. exitation, magn. field strength (”magn. displacement”)

$n =$ spacial molecule density in volume

$\alpha = 4\pi r^3 = constant \quad \chi =$ elec. susceptibility of dieelectricum $\quad \kappa = \dfrac{\mu - \mu_0}{\mu_0} =$ magn. susceptibility

$\langle p_m \rangle =$ spacial average of magn. dipol moment $\quad \langle \mathcal{E}_m \rangle =$ spacial average of electric field

$\bar{\mathcal{E}}_m =$ time average of electric field $\quad \mu =$ magn. permeability $\quad \mu_0 =$ magn. constant ...

$\varepsilon =$ dielectricity constant $\quad \varepsilon_0 =$ ...

Electro-Magnetic interactions/coupling (Maxwell conditions)

conservativ, static charge localisation:

...

stationary, slow charge movement:

$$\nabla \cdot \varepsilon_0 \mathcal{E} = \varrho \qquad \nabla \cdot \mathcal{B} = 0 \quad \nabla \times \mathcal{E} = -\frac{\partial}{\partial t}\mathcal{B} = i\omega\mathcal{B} \qquad \nabla \times \frac{1}{\mu_0}\mathcal{B} = j + \varepsilon_0 \frac{\partial}{\partial t}\mathcal{E} = j - i\omega\varepsilon_0\mathcal{E} \tag{5}$$

relativistic, fast charge movement:

...

Potential

$\Rightarrow$ Wave equation that determines Potential:

$$\Delta \cdot U - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}U = J \tag{6}$$

$\Rightarrow$ Potential caused by "ignition J" (inhomgenity) from source at rest, v = 0:

$$U = -\frac{1}{4\pi}\int \frac{J}{|r|}\,dr' \tag{7}$$

elec. Potential structure:

$J = -\frac{1}{\varepsilon_0}\varrho$

$$U_e := \Phi \overset{\text{elec. charge dens.}=\varrho}{=} \frac{1}{4\pi}\frac{1}{\varepsilon_0}\int \frac{\varrho}{|r|}\,dr' \mathrel{\hat{=}} \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\frac{1}{\varepsilon_0}Q = \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\frac{1}{\varepsilon_0}It \tag{8}$$

magn. Potential structure:

$J = -\mu_0 j$

$$U_m := A \overset{\text{elec. current dens.}=j}{=} \frac{\mu_0}{4\pi}\int \frac{j}{|r|}\,dr' \mathrel{\hat{=}} \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\mu_0 I = \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\frac{1}{c^2\varepsilon_0}I = \frac{1}{c^2}\frac{1}{t}U_e \overset{\text{source in uniformal movement},v=const}{=} \tag{9}$$

Electric endogen energy

- Bond Energy
- Binding Energy

$E_{pot} = mc^2$

$$e = \text{electron charge} \qquad q = ne = \text{charge} \qquad n \in \mathbb{N} \tag{10}$$

$$\text{Potential} \qquad U = \frac{E_{pot}}{q} = \frac{\left(mc^2\right)^2}{ne} = \frac{m}{q}mc^4 = \kappa mc^4 \tag{11}$$

Electric induced energy (exogenous excitation)

- Ionisation Energy
- Exitation Energy

$E_{kin} = \frac{1}{2}mv^2 \mathrel{\hat{=}} (pc)^2 \mathrel{\hat{=}} h\nu$

E_{kin} Energy density electric:

$$w_e = \frac{1}{2}\varepsilon_0 \mathcal{E}^2 \tag{12}$$

E_{kin} Energy density magnetic:

$$w_m = \frac{1}{2}\frac{1}{\mu_0}\mathcal{B}^2 \tag{13}$$

E_{kin} Electric Energy:

$$W_e = \frac{1}{2}CU^2 \tag{14}$$

E_{kin} Magnetic Energy:

$$W_m = \frac{1}{2}LI^2 \tag{15}$$

Power:

$$P = VI = (U_1 - U_2)\frac{Q}{t} = \frac{E_1 - E_2}{t} = \frac{W}{t} = \frac{Fr}{t} = Fv \tag{16}$$

Conductor (charge conductive element for energy transfer) condition:

$$\sigma \cdot t \gg \varepsilon_0 \tag{17}$$

Slow changing current (quasi-stationary, low frequencies) condition:

$$\frac{v}{c} \ll \frac{1}{c}\frac{\mathcal{E}}{\mathcal{B}} \ll \frac{c}{v} \tag{18}$$

Energy conservation for (large extended) conductive medium:

$$\int \left(\frac{\partial}{\partial t} (we + w_m) + \ \cdot \mathcal{S} + j \cdot \mathcal{E} \right) dr = \frac{d}{dt} W_e + \frac{d}{dt} W_m + D_S + \int (j \cdot \mathcal{E}) \ dr \overset{\text{low frequency}}{=} \frac{1}{2} \frac{d}{dt} (LI^2) + \frac{1}{2} \frac{d}{dt} \left(\frac{Q^2}{C} \right) \tag{19}$$

Energy of thin conductor (concentrated conductive medium without electric charge in its surrounding), Voltage (potentialdifference, tension):

$$V = U_1 - U_2 = r\mathcal{E} = \dot{\Psi} + RI + \frac{I}{C} t = L\dot{I} + RI + \frac{Q}{C} \tag{20}$$

$$\Delta \cdot := \text{ div grad} = \text{Laplacian}$$

$$J = \text{"ignition of inhomogenity,cause"} \quad j = \text{Current density} \quad \varrho = \text{Charge density}$$

$$V = \text{Potentialdifference} \quad I = \text{Current} \quad R = \text{Resistance}$$

$$L = \text{Inductance} \qquad C = \text{Capacitance} \qquad Q = \text{charge (ion)} \qquad \Psi = LI = \text{mag. Flux}$$

$$\mathcal{S} = \mathcal{E} \times \mathcal{H} = \text{Poynting-Vector} = \text{radiation intensity}$$

$$D_S = \int (\ \cdot \mathcal{S}) \ dr = \int \text{div}(\mathcal{E} \times \mathcal{H}) \, dr = \text{dissipation engergy lost from radiation} \approx 0 \text{ for low frequencies } \omega$$

$$d = \sqrt{\frac{1}{\mu\sigma} \frac{2}{\omega}} = \text{skin depth (surface) of conductor} \qquad r = \frac{d}{\sqrt{2i}} \rho = \frac{d}{1+i} \rho = \text{radius (extension) of conductor}$$

$$\rho = kr = \text{"radius of curvature } k\text{"}$$

$$\omega = \dot{\rho} = \frac{v}{r} = 2\pi\nu = k\lambda \frac{1}{T} = \sqrt{\frac{k}{m}} \mathrel{\hat{=}} kc \mathrel{\hat{=}} \frac{1}{\sqrt{LC}} \mathrel{\hat{=}} \sqrt{\frac{g}{r}} = \text{frequency}$$

μ = permeability of conductor $\quad$ σ = conductivity of conductor (dielectricum)

Low Frequency

$$\omega \ll \frac{1}{\mu\sigma}\frac{2}{r^2} \iff r \ll d \iff \rho \ll 1 \tag{21}$$

$$L \approx const. \qquad C \approx const. \qquad D_S \approx 0$$

High Frequency

$$\omega \gg \frac{1}{\mu\sigma}\frac{2}{r^2} \iff r \gg d \iff \rho \gg 1 \tag{22}$$

$$L(t) \neq const. \qquad C(t) \neq const. \qquad \text{for Dipol: } D_S \propto \omega^4 \gg 0$$

Continuity at Rest: Electic Charge

no Dissipation, no velocity, charge (ion) causes Electric Field

Continuity at Movement: Magnetic Field

no Dissipation, Velocity constant, moved charge (ion) causes emergence of Magnetic Field

Continuity at Acceleration: Radiation Energy

no Dissipation, Velocity changing, acceleration of charge (ion) causes emgergence of Radiation

Continuity and periodic (harmoneous oscillation) Movement in VACUUM

$$2\pi = k\lambda \gg kr_0 \ll 1$$

$$\omega = kc \ll \frac{c}{r_0} \qquad \nabla \cdot j = i\omega\varrho \qquad i\omega Q = 0$$

Continuity and periodic (harmoneous oscillation) Movement in MATTER

Refraction index emerges

Dispersion: $k^2 = \varepsilon\mu\omega^2 + i\mu\sigma\omega$

Near Field

$$r \ll \lambda = \frac{2\pi}{k} \overset{\text{vacuum}}{=} \frac{2\pi}{\frac{\omega}{c}} \qquad kr \ll 2\pi$$

$$\frac{w_m}{w_e} \sim \frac{\frac{\vec{p}^2}{\varepsilon_0 r^6}}{\frac{k^2\mu_0 c^2 \vec{p}^2}{r^4}} \sim (kr)^2 \ll 1$$

$$\vec{p} = \text{elec. Dipol}$$

Far Field

accelerated movement -> Radiation in Far Field of charge source

$$r \gg \lambda = \frac{2\pi}{k} \overset{\text{vacuum}}{=} \frac{2\pi}{\frac{\omega}{c}} \qquad kr \gg 1 \qquad r \gg kr_0^2 \ll 1$$

$$\frac{w_m}{w_e} = \frac{\frac{\varepsilon_0}{2} c^2 \left(\mathcal{B} \times \frac{r}{|r|}\right)^2}{\frac{\varepsilon_0}{2} c^2 \left(\mathcal{B} \times \frac{r}{|r|}\right)^2} = 1$$

Dissipation in Vacuum

no Contiunuity of chrge/matter, no Refraction index, Reflection

Dissipation in Matter

no Contiunuity of chrge/matter with Dielectricum, Refraction merges, Reflection, Absorbtion

Energy gain by excitation:

Action Quantum, Planck Constant (Energy-Frequency-Slope)[1]:

$$h = \tan(\alpha) = \frac{E_{kin}}{\nu_i - \nu_0} = \frac{eV_{0i}}{\nu_i - \nu_0} \equiv \\ \frac{E_1 - E_2}{\nu_0} = \frac{W}{\nu_0} = \frac{F_{EM} \cdot x}{\nu_0} = \hbar k \lambda \tag{23}$$

$$\hbar = \frac{E_1 - E_2}{\omega} = \tag{24}$$

$$E_{kin}^{max} = eV_0 = e\frac{Q}{C} = h(\nu - \nu_0) \tag{25}$$

$$V_0 = \text{Opposing Potential } (I = 0)$$

$$\alpha = \angle(E_{kin}, \nu) \qquad e = \text{electron charge} \qquad Q = \text{ion charge}$$

$$C = \text{Capacitance} \qquad \nu = \frac{1}{2\pi}\frac{1}{\sqrt{LC}}\sqrt{1 - \frac{C}{L}R^2}$$

$$\Rightarrow \text{Inductance } L = \frac{4\pi^2}{C}\left(\frac{e}{h}Q + \nu_0\right)^2$$

$\Rightarrow$ Ionisation Energy of chemical element $E_i = h\nu = h\frac{\lambda}{c} = \hbar\omega = \frac{h}{k\lambda}\omega$

$\Rightarrow$ Energy released (gained) $E_{gain} = E_{pot} - E_i = mc^2 - h\nu$

[1] Wirkungsquantum

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